

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

- The Delta tables for customer and order data do not exist or are empty.

Main Flow / Functional Steps:

1. Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and load it into a Delta table named "customer".
2. Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` and load it into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" table, calculated as the product of "PricePerUnit" and "Qty".
4. Remove null and duplicate records from both "customer" and "order" tables.

FR-DLT-002: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined data from "customer" and "order" tables using SCD Type 2.

Preconditions:

- "customer" and "order" tables are loaded with data.
- "ordersummary" table does not exist or is empty.

Main Flow / Functional Steps:

1. Create "ordersummary" table if it does not exist in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard".

2. Join "customer" and "order" tables on "CustId" field.
3. Load the joined data into "ordersummary" table using SCD Type 2.

FR-DLT-003: Update ordersummary Table on Customer Change

Title: Update ordersummary Table on Customer Change

Description: The system must update the "ordersummary" table whenever there is a change in the "customer" table.

Preconditions:

- "customer" table is updated.
- "ordersummary" table exists and is loaded with data.

Main Flow / Functional Steps:

1. Identify changes in the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update StartDate and EndDate accordingly to maintain history.

FR-DLT-004: Create and Load customeraggregatespend Table

Title: Create and Load customeraggregatespend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load it with aggregated data from "ordersummary" table.

Preconditions:

- "ordersummary" table exists and is loaded with data.
- "customeraggregatespend" table does not exist or is empty.

Main Flow / Functional Steps:

1. Create "customeraggregatespend" table if it does not exist in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard".
2. Aggregate "TotalAmount" column from "ordersummary" table and group by "Name" and "Date" columns.
3. Load the aggregated data into "customeraggregatespend" table.

FR-DLT-005: Implement using Delta Live Tables

Title: Implement using Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements are implemented.

Main Flow / Functional Steps:

1. Implement steps 1 to 4 using Delta Live Tables.